Dinesh

Senior Data Engineer

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PROFESSIONAL SUMMARY

Highly skilled Senior Data Engineer with a demonstrated track record of successfully delivering end-to-end data engineering solutions. Equipped with overall **5+ years** of experience in architecting and implementing scalable data pipelines and optimizingdata infrastructure for improved performance and efficiency.

- **Data Engineering**: Proficient in designing and implementing robust data engineering solutions using technologies such as Apache Spark, Apache Kafka, Hadoop, and cloud-based data services. Solid understanding of data ingestion, data transformation, data modeling techniques and data visualization.
- **Data Architecture**: Extensive experience in designing and implementing scalable data architectures, including data lakes, data warehouses, and real-time streaming platforms. Expert in data modeling techniques and optimizing data structures for efficient querying and analytics.
- **ETL Development**: Expertise in building end-to-end Extract, Transform, Load (ETL) processes to integrate and transform data from various sources. AWS or custom Python-based solutions for data integration and processing.
- **Big Data Technologies**: Strong knowledge of distributed computing frameworks like Hadoop and Spark, along with relatedtools such as Hive, Pig, and HBase. Experienced in working with large-scale data sets and implementing efficient data processing workflows.
- **Programming and Scripting**: Proficient in programming languages like Python, R, Java, and Scala. Skilled in writing complexSQL queries for data manipulation and retrieval. Familiarity with scripting languages like Bash or PowerShell for automationand orchestration tasks.
- **Cloud Data Platforms**: Hands-on experience with cloud-based data platforms such as AWS, GCP, or Azure. Seasoned in leveraging cloud services like Amazon S3, Google Cloud Storage, or Azure Data Lake Store for data storage and processing.Experience with cloud-native data services like AWS Glue, GCP Dataflow, or Azure Databricks.
- **Data Quality and Governance**: Strong understanding of data quality and data governance principles. Competent in implementing data validation checks, data lineage tracking, and data cataloging. Familiarity with data privacy regulations and best practices for ensuring data security using Collibra and Alation.
- **Team Collaboration**: Excellent communication and interpersonal skills, with a proven ability to collaborate effectively withcross-functional teams and stakeholders using Agile methodology. Skilled in leading and mentoring junior data engineers, fostering a collaborative and productive work environment.

EDUCATION

- **Masters in Technology Management from Lindsey Wilson College.**

TECHNICAL SKILLS

Languages	Python, R, SQL, Java, C#, Unix Shell Scripting, Scala
Databases	MySQL, SQL Server, Oracle, Postgres, Mongo DB, Neo4j, Cassandra, IBM Db2, Azure Databricks
Hadoop Ecosystem	Spark, Hadoop, HDFS, Hive, Kafka, Pyspark, Yarn
Data Engineering Tools **Data Warehouse Tools** **Data Governance**	Airflow, Dremio, Alation, Infogix, Looker, Talend Snowflake, Data Bricks, Big Query, Redshift, Db2 Collibra, Alation
Cloud services	AWS- EC2, EMR, S3, RDS, IAM, ECS, SQS, QuickSight Azure Maria DB, CloudWatch, CloudFormation, Azure Cosmo DB, Microsoft Azure Data Factory, Virtual machines, Blob storage, GCP
Versioning Tools	GIT, Bitbucket, GitHub
Operating systems	Linux/Unix, Ubuntu, Windows,
Scrum Methodologies	Agile methodology, Waterfall model

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Progressive Insurance, Overland Park, KS **Dec 2022 to Present**

Responsibilities:

- Oversaw end-to-end operations of ETL data pipelines in Azure Data Factory, maintaining high-quality data integration, transformation, and loading processes.
- Implemented Logic Apps to streamline order processing workflows and enhance operational efficiency in the insurance industry, resulting in improved order fulfilment and streamlined data exchange.
- Developed and implemented data ingestion pipelines using Azure Synapse Analytics to efficiently extract, transform, and load (ETL) large volumes of structured and unstructured data into the data warehouse.
- Designed and optimized data models and schemas within Azure Synapse Analytics to ensure efficient storage, retrieval, and querying of data, resulting in improved performance and reduced costs.
- Leveraged Snowflake to optimize data warehousing and analytics capabilities in the insurance industry, enabling valuable insights, data-driven decision-making, and improved targeted marketing campaigns.
- Deployed Azure Event Hub for real-time data ingestion in the insurance industry, facilitating real-time monitoring of foot traffic, enhanced fraud detection, and improved operational agility.
- Managed the orchestration of Terraform for infrastructure automation in Azure deployments, streamlining provisioning and management of resources.
- Utilized Ansible for configuration management, automating infrastructure provisioning and ensuring consistency across environments.
- Utilized Hive within the Azure ecosystem to derive valuable insights, optimize queries, enable efficient data warehousing, and deliver comprehensive insurance analytics as an Azure Data Engineer, enabling data-driven decision-making and enhancing business performance.
- Employed PySpark and Scala in conjunction with Azure HDInsight, Kafka, and Spark Streaming to enable advanced data processing, real-time streaming analytics, and scalable data solutions within the Azure ecosystem, driving actionable insights and enhancing decision-making capabilities in the insurance industry.
- Implemented and managed containerization using Docker, enhancing application scalability and portability.
- Developed and maintained Java-based applications using Spring Boot, ensuring high performance and reliability.
- Effectively utilized Oozie and Zookeeper for orchestration and coordination of data workflows in Azure, optimizing storage efficiency with ORC, Avro, Parquet, and delimited file formats as part of Azure data solutions, ensuring seamless data processing and efficient data storage in the insurance data engineering landscape.
- Collaborated with DevOps teams to orchestrate and manage containerized applications on Kubernetes, optimizing resource utilization.
- Ensured data security through encryption while seamlessly handling data ingestion, protecting sensitive data and maintaining data integrity in compliance with industry regulations in the insurance industry.
- Supervised the development and deployment of SSIS/SSRS packages for improved data accuracy and business insights in the insurance industry, enabling data-driven decision-making, generating comprehensive reports, and facilitating data visualization.
- Prepared technical specifications, data flow diagrams, and process documentation, fostering clear communication and effective collaboration within the team.

Tools & Technologies: Azure Databricks, Data Factory, Logic Apps, EventHub, Spark Streaming, Data pipeline, Terraform, Azure DevOps, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, Kafka, Power BI.

Data Engineer

CNA Insurance, Chicago, IL **Oct 2020 to Nov 2022**

Responsibilities:

- Managed the development and execution of ETL data pipelines in Azure Data Factory, improving the processing of millions of auctions transactions daily and facilitating faster auctions reporting.
- Administered Azure Data Lake and Azure SQL Database for optimal data storage solutions, preserving customer data for over 10 million accounts and ensuring high availability.
- Leveraged Azure Databricks and HDInsight for Big Data processing, enhancing credit risk modelling and customer segmentation, and driving actionable insights for data-based decision making.

- Implemented robust data security measures using Azure Security Center and Azure Key Vault, protecting sensitive auctions information, and ensuring compliance with GDPR and other auctions industry regulations.
- Orchestrated automated loan approval workflows using Azure Logic Apps, reducing processing time and enhancing customer experience.
- Worked with Azure DevOps to streamline CI/CD pipelines, enabling faster updates to customer-facing banking applications and reducing time-to-market for software releases.
- Streamlined deployment and scalability of data processing environments using Docker containerization and Azure.
- Implemented real-time data processing using Spark Streaming and PySpark, enabling rapid insights from transaction data and improving fraud detection and risk management.
- Enhanced data quality and integrity by ingesting and transforming data from diverse sources, and developed ETL processes using SSIS and SSRS packages to facilitate data-driven decision-making.
- Played a pivotal role in cross-functional teams to develop data models and solutions that drove a 20% improvement in meeting business needs.
- Implemented robust data governance and security measures, safeguarding sensitive customer data, and ensuring compliance with banking industry regulations.
- Boosted system efficiency by optimizing data pipelines and queries, and improved speed of generating banking reports using efficient Hive queries.
- Enhanced data processing efficiency by working extensively with RDDs, Data frames, and Hive for data analysis and processing, and optimized batch processing of streaming data with Spark Streaming.

Tools & Technologies: Azure Databricks, Data Factory, Logic Apps, EventHub, Spark Streaming, Terraform, Azure DevOps, YAML, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, Kafka, Power BI.

Data Warehouse Developer
Bank of America, Plano, TX **Feb 2019 to Oct 2020**
Responsibilities:

- Developed stored procedures, triggers, and functions, optimizing SQL Server performance through indexing and monitoring techniques.
- Led the origination and execution of debt and equity capital market transactions, including bond issuances, initial public offerings (IPOs), and secondary offerings, by collaborating with internal teams and external stakeholders to ensure successful deal execution.
- Implemented ETL data flows using SSIS, facilitating data migration and transformation from various sources including SQL Server, Access, and Excel.
- Managed client relationships and provided tailored capital market solutions to corporate clients, advising on optimal funding structures and capital raising strategies to support their growth objectives and enhance shareholder value
- Gained experience in dimensional data modelling and identified facts and dimensions for Data Mart design, along with developing fact tables and dimension tables using Slowly Changing Dimensions (SCD) techniques.
- Managed error and event occurrences during ETL processes, employing techniques such as precedence constraints, breakpoints, checkpoints, and logging.
- Developed and implemented risk management frameworks for capital market activities, including credit risk assessment, market risk monitoring, and liquidity management, to ensure compliance with regulatory requirements and safeguard the bank's financial stability
- Acquired experience in building SSAS cubes, implementing aggregations, defining KPIs, partitioning cubes, and creating data mining models, contributing to deployment, and processing of SSAS objects.
- Developed a range of reports, including parameterized, chart, graph, linked, dashboard, scorecards, and drill-down/drill-through reports on SSAS cubes using SSRS.
- Utilized advanced financial modeling skills to structure and price capital market products, including structured notes, hybrid securities, and derivative-based investment solutions, to meet the unique risk-return preferences of institutional and retail clients.

Tools & Technologies: MS SQL Server, Visual Studio Legacy Versions, SSIS, Share point, MS Access, Git.